

Conical surface deflector

As per newton's second law of motion, the force of the jet strikes a plate is equal to the rate of change of momentum of the jet, therefore; For the Conical deflector ($\alpha = 0^\circ$, $\beta = 135^\circ$):

$$F = \text{Mass of water flowing} \times \text{Change of velocity}$$

So,

$$F = \rho A V \times \Delta V = \rho A V \times (V \cos(\alpha) - V \cos(\beta))$$

Therefore:

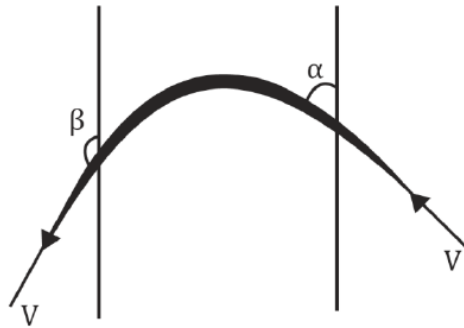
$$F = 1.707 \rho A V^2$$

where:

$$V^2 = V_0^2 - 2gS$$

and,

$$V_0 = Q/A$$



Where:

F = the force exerted by the jet in N.

ρ = density of water in kg/m^3 .

A = cross sectional area of the jet in m^2 .

V = actual velocity at the center of the jet in m/s .

ΔV = change of the velocity in m/s .

V_0 = velocity of the jet in m/s .

S = the distant between the nozzle opening and the deflector opening in m.

g = is the gravitational acceleration in m/s^2 .

Q = the flow rate in m^3/s .

Answer:

$$A = 0.0000785 \text{ m}^2, S = 0.035 \text{ m}, g = 9.81 \text{ m/s}^2, \rho = 1000 \text{ kg/m}^3$$

❖ $Q = \text{-----} \text{ m}^3/\text{s}.$

❖ $V_o = Q / A = \text{-----} = \text{-----} \text{ m/s}.$

❖ $V^2 = V_o^2 - 2gS = \text{-----} = \text{-----} = \text{m/s}.$

❖ $F_{th} = 1.707 \rho A V^2 = \text{-----} = \text{-----} \text{ kg.m/s}^2 \text{ (N)}$

❖ **Fac.** = total weight = total mass $\times$ g = ----- kg.m/s² (N)

❖ Plot the graph between **F_{th}.**& **F_{ac.}** vs. **Q**:

Q (L/min.)									
F (theoretical)									
F (actual)									

